

| Question |     |      |    | Marking details                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) |      |    | 7                                                                                                                                                                                                                                      |                 | 1   |     | 1     |       |      |
|          | (b) |      |    | Cl <sub>2</sub>                                                                                                                                                                                                                        |                 | 1   |     | 1     |       |      |
|          | (c) | (i)  |    | glowed less brightly than iodine <input type="checkbox"/><br>glowed less brightly than chlorine <input checked="" type="checkbox"/><br>glowed more brightly than chlorine <input type="checkbox"/>                                     |                 | 1   |     | 1     |       | 1    |
|          |     | (ii) | I  | FeBr <sub>3</sub>                                                                                                                                                                                                                      |                 | 1   |     | 1     |       |      |
|          |     |      | II | iron bromide ignore any bracketed numbers                                                                                                                                                                                              |                 | 1   |     | 1     |       |      |
|          | (d) |      |    | to disinfect skin before surgery <input checked="" type="checkbox"/><br>to make coloured fireworks <input type="checkbox"/><br>to sterilise swimming pools <input type="checkbox"/><br>to fill party balloons <input type="checkbox"/> | 1               |     |     | 1     |       |      |
|          |     |      |    | Question 3 total                                                                                                                                                                                                                       | 1               | 5   | 0   | 6     | 0     | 1    |